CSAI2121
Probability and Statistics

Integrated Exercise Set

Chapters 1–4

Total: 30 Exercises

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Part I Conditional Probability and Bayes' Theorem

Exercise 1: Evaluating a Classifier

A binary classifier is tested on 10,000 observations. Among these observations, 800 are actually positive and 9,200 are actually negative.

The classifier produces the following results:

	Predicted Positive	Predicted Negative
Actually Positive	680	120
Actually Negative	460	8740

Let D denote the event that an observation is actually positive and let T denote the event that the classifier predicts positive.

(a) Find $P(D)$, $P(T)$, and $P(D \cap T)$.

(b) Find the precision $P(D \mid T)$.

(c) Find the recall, or sensitivity, $P(T \mid D)$.

(d) Find the specificity $P(T^c \mid D^c)$.

(e) Find the false omission probability $P(D \mid T^c)$.

(f) Determine whether D and T are independent. Justify your answer.

Exercise 2: Sampling Defective Chips Without Replacement

A container contains 12 computer chips, of which 3 are defective and 9 are non-defective. Three chips are selected uniformly at random without replacement.

Let A_i denote the event that the i-th selected chip is defective.

(a) Find the probability that exactly one selected chip is defective.

(b) Find the probability that at least one selected chip is defective.

(c) Find $P(A_1)$.

(d) Find $P(A_2 \mid A_1)$.

(e) Find $P(A_1 \mid A_1 \cup A_2 \cup A_3)$.

(f) Are A_1 and A_2 independent? Explain.

Exercise 3: Identifying the Source of a System Error

A company uses three machine-learning models to process incoming requests. The proportions of requests handled by the models are

$$P(M_1) = 0.50, \qquad P(M_2) = 0.30, \qquad P(M_3) = 0.20.$$

Their respective error probabilities are

$$P(E \mid M_1) = 0.02, \qquad P(E \mid M_2) = 0.04, \qquad P(E \mid M_3) = 0.08,$$

where E denotes the event that an error occurs.

(a) Find the overall probability that a randomly selected request produces an error.
(b) Given that an error occurred, find $P(M_1 \mid E)$.
(c) Given that an error occurred, find $P(M_2 \mid E)$.
(d) Given that an error occurred, find $P(M_3 \mid E)$.
(e) Which model is the most likely source of an observed error?
(f) Explain why the model with the largest error rate is not necessarily responsible for most of the observed errors.

Exercise 4: Two Diagnostic Tests

A disease occurs in 2% of a population. Two diagnostic tests are used.

For a person who has the disease,

$$P(T_1^+ \mid D) = 0.90, \qquad P(T_2^+ \mid D) = 0.95.$$

For a person who does not have the disease,

$$P(T_1^+ \mid D^c) = 0.08, \qquad P(T_2^+ \mid D^c) = 0.05.$$

Assume that the two test results are conditionally independent given the person's true disease status.

(a) Find $P(T_1^+ \cap T_2^+ \mid D)$.
(b) Find $P(T_1^+ \cap T_2^+ \mid D^c)$.
(c) Find $P(D \mid T_1^+ \cap T_2^+)$.
(d) Find the probability of obtaining exactly one positive test result, given D.
(e) Find the probability of obtaining exactly one positive test result, given D^c.
(f) Find the posterior probability of disease given exactly one positive test result.
(g) Compare the evidence provided by two positive results with the evidence provided by exactly one positive result.

Exercise 5: Independence from a Joint Probability Table

The joint distribution of two discrete random variables X and Y is given by

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.10	0.15	0.05
$X = 1$	0.20	0.30	0.20

Define the events

$$A = \{X = 1\}, \qquad B = \{Y \geq 1\}, \qquad C = \{Y = 0\}.$$

(a) Find the marginal distribution of X.

(b) Find the marginal distribution of Y.

(c) Find $P(A)$, $P(B)$, and $P(A \cap B)$.

(d) Determine whether A and B are independent.

(e) Determine whether B and C are mutually exclusive.

(f) Find $P(A \mid B)$.

(g) Find $P(A \cup B)$.

Exercise 6: A Two-Stage Cloud-Routing System

A computing job is initially routed to one of two cloud regions:

$$P(E) = 0.60, \qquad P(W) = 0.40,$$

where E represents the East region and W represents the West region.

The probabilities that the first attempt succeeds are

$$P(S_1 \mid E) = 0.95, \qquad P(S_1 \mid W) = 0.90.$$

If the first attempt fails, the job is rerouted to the other region. Assume that the success or failure of the second attempt is conditionally independent of the first attempt, given the region handling that attempt.

(a) Find the probability that the first attempt fails.

(b) Given that the first attempt failed, find the probability that the job was initially routed to the West region.

(c) Find the probability that the first attempt fails but the second attempt succeeds.

(d) Find the probability that both attempts fail.

(e) Find the probability that the job eventually succeeds.

(f) Given that both attempts failed, find the probability that the job was initially routed to the East region.

(g) Explain why the answer in part (b) differs from $P(W) = 0.40$.

Part II Discrete Random Variables

Exercise 7: Finding and Analysing a PMF

A discrete random variable X has probability mass function

$$p_X(x) = \begin{cases} c(x+1), & x = 0, 1, 2, 3, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the normalization constant c.
(b) Write the complete probability distribution of X.
(c) Find $P(X \geq 2)$.
(d) Find $P(1 \leq X < 3)$.
(e) Find $E[X]$.
(f) Find $E[X^2]$.
(g) Find $\text{Var}(X)$ and $\text{SD}(X)$.

Exercise 8: Recovering a PMF from a CDF

The cumulative distribution function of X is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.15, & 0 \leq x < 1, \\ 0.45, & 1 \leq x < 2, \\ 0.80, & 2 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

(a) Determine the possible values of X.
(b) Find the PMF $p_X(x)$.
(c) Find $P(X < 2)$.
(d) Find $P(1 \leq X \leq 4)$.
(e) Find $P(X > 1 \mid X \leq 4)$.
(f) Find $E[X]$.
(g) Find $\text{Var}(X)$.

Exercise 9: Expected Errors Using Indicator Variables

A software system processes 200 independent requests. Request i produces an error with probability p_i.

For the first 100 requests, $p_i = 0.01$, while for the remaining 100 requests, $p_i = 0.03$.

Let

$$I_i = \begin{cases} 1, & \text{if request } i \text{ produces an error,} \\ 0, & \text{otherwise,} \end{cases}$$

and let

$$X = \sum_{i=1}^{200} I_i.$$

(a) Find $E[I_i]$ for each of the two groups.
(b) Use indicator variables to find $E[X]$.
(c) If every error costs the company \$50, find the expected total cost.
(d) Assuming independence, find $\mathrm{Var}(X)$.
(e) Is independence required in order to calculate $E[X]$? Explain.
(f) Suppose each error in the second group costs \$80 instead of \$50. Find the expected total cost under this new cost structure.

Exercise 10: A Binomial Tail Probability

Each message transmitted through a network is corrupted independently with probability 0.08. Let X be the number of corrupted messages among 20 transmitted messages.

(a) Identify the distribution of X and state its parameters.
(b) Find $P(X = 2)$.
(c) Find $P(X \geq 3)$.
(d) Find $P(X \leq 1)$.
(e) Find $P(X \geq 3 \mid X \geq 1)$.
(f) Find $E[X]$ and $\mathrm{Var}(X)$.
(g) Determine the mode of X.

Exercise 11: Scaling a Poisson Rate

Requests arrive at a server according to a Poisson process with an average rate of 6 requests per minute.

(a) State the Poisson parameter for a 30-second interval.
(b) Find the probability of exactly 4 arrivals in 30 seconds.
(c) Find the probability of at most 2 arrivals in 30 seconds.
(d) Find the probability of no arrivals in 10 seconds.
(e) Find the probability of at least one arrival in 10 seconds.
(f) Find the expected number and variance of arrivals in 5 minutes.
(g) Find the probability of more than 35 arrivals in 5 minutes.

Exercise 12: Transforming a Poisson Random Variable

Let

$$X \sim \text{Poisson}(4)$$

represent the number of system failures observed during a fixed period. The maintenance cost, measured in hundreds of dollars, is

$$Y = 3X + 2.$$

(a) List the first five possible values of Y.
(b) Find $P(Y = 11)$.
(c) Find $P(Y \leq 8)$.
(d) Find $E[Y]$.
(e) Find $\text{Var}(Y)$.
(f) Find $\text{SD}(Y)$.
(g) Is Y a Poisson random variable? Explain.

Exercise 13: Selecting and Critiquing Probability Models

For each situation below:

- identify an appropriate probability distribution;
- state its parameter or parameters;
- describe one important assumption required by the model.

(a) Whether a single online payment succeeds.
(b) The number of successful payments among 50 independent payments with the same success probability.
(c) The number of customer arrivals during a fixed time interval.
(d) A randomly selected time between 2:00 p.m. and 2:30 p.m., assuming all times are equally likely.
(e) The waiting time until the next customer arrival in a Poisson process.
(f) The height of an adult selected from a large population.
(g) The number of defective items obtained when 10 items are sampled without replacement from a batch containing 8 defective and 92 non-defective items.

Exercise 14: Waiting for the First Failed Transaction

A transaction fails independently with probability 0.15. Let X denote the number of transactions processed up to and including the first failed transaction.

Thus, $X = 1$ means that the first transaction fails.

(a) Identify the distribution of X and state its parameter.

(b) Write down the PMF of X.

(c) Find $P(X = 4)$.

(d) Find $P(X > 5)$.

(e) Find $P(X \leq 8)$.

(f) Find $P(X > 12 \mid X > 5)$.

(g) Find $E[X]$.

(h) Find $\text{Var}(X)$.

(i) Explain the memoryless property represented by part (f).

Part III Continuous Random Variables

Exercise 15: A Quadratic Probability Density

A continuous random variable X has PDF

$$f_X(x) = \begin{cases} cx^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of c.
(b) Find the CDF $F_X(x)$.
(c) Find $P(0.5 < X < 1.5)$.
(d) Find $P(X \ge 1)$.
(e) Find $E[X]$.
(f) Find $E[X^2]$.
(g) Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

Exercise 16: CDF, Quantiles, and Conditional Probability

The CDF of a continuous random variable X is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - (1-x)^3, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

(a) Find the PDF of X.
(b) Find $P(0.2 < X < 0.7)$.
(c) Find $P(X > 0.6 \mid X > 0.3)$.
(d) Find the median of X.
(e) Find the 90th percentile of X.
(f) Find $E[X]$.
(g) Compare the mean and median. What does the comparison suggest about the shape of the distribution?

Exercise 17: Uniformly Distributed Network Latency

The latency X, measured in milliseconds, is uniformly distributed between 20 and 80:

$$X \sim \mathrm{Uniform}(20, 80).$$

(a) Write down the PDF and CDF of X.
(b) Find $E[X]$.

(c) Find $\mathrm{Var}(X)$.

(d) Find $P(X < 35)$.

(e) Find $P(X > 60 \mid X > 40)$.

(f) Find the 95th percentile of X.

(g) Find an interval centred at the mean that contains 80% of the probability.

Exercise 18: Exponential Waiting Time and Memorylessness

The waiting time X, measured in minutes, until the next request arrives has an exponential distribution with rate

$$\lambda = 0.25.$$

(a) Write down the PDF and CDF of X.

(b) Find the expected waiting time.

(c) Find the variance of the waiting time.

(d) Find $P(X < 3)$.

(e) Find $P(2 < X < 6)$.

(f) Find $P(X > 10 \mid X > 4)$.

(g) Find the 90th percentile of X.

(h) Explain how part (f) illustrates the memoryless property.

Exercise 19: Exam Scores and Normal Probabilities

Exam scores are normally distributed with mean 70 and standard deviation 10:

$$X \sim N(70, 10^2).$$

(a) Find the z-score corresponding to a score of 85.

(b) Find $P(X < 60)$.

(c) Find $P(65 < X < 85)$.

(d) Find the score that separates the highest 10% of students from the remaining students.

(e) Find the central interval containing 95% of the scores.

(f) A student receives a score of 42. Is this score unusual under the 2-standard-deviation rule? Explain.

Exercise 20: Transforming a Confidence Score

A confidence score X has PDF

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

A transformed score is defined by

$$Y = 100X + 10.$$

(a) Find $E[X]$ and $\mathrm{Var}(X)$.
(b) Find $E[Y]$.
(c) Find $\mathrm{Var}(Y)$.
(d) Find $P(Y > 80)$.
(e) Determine the range of possible values of Y.
(f) Find the CDF $F_Y(y)$.
(g) Find the PDF $f_Y(y)$.

Exercise 21: Comparing Two Response-Time Models

Two response-time models have the same mean of 5 seconds.

Model A is uniform:

$$X \sim \mathrm{Uniform}(2, 8).$$

Model B is exponential:

$$Y \sim \mathrm{Exponential}(0.2).$$

(a) Verify that both models have mean 5.
(b) Find $\mathrm{Var}(X)$ and $\mathrm{Var}(Y)$.
(c) Find $P(X > 7)$.
(d) Find $P(Y > 7)$.
(e) Find $P(X > 10)$ and $P(Y > 10)$.
(f) Which model has the larger tail probability for long response times?
(g) Discuss which model would be more appropriate if response times can never exceed 8 seconds.

Exercise 22: Recovering a PDF from a Nonstandard CDF

The CDF of a random variable X is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \dfrac{x^2}{1 + x^2}, & x \ge 0. \end{cases}$$

(a) Verify that $F_X(x)$ satisfies the main properties of a CDF.

(b) Find the PDF $f_X(x)$.

(c) Find $P(1 < X < 3)$.

(d) Find $P(X > 2)$.

(e) Find the median of X.

(f) Determine whether $E[X]$ is finite. Justify your answer using an appropriate integral or tail argument.

(g) Based on part (f), discuss whether $\mathrm{Var}(X)$ can be finite.

Exercise 23: A Decreasing Continuous Density

Suppose that the lifetime X, measured in hours, has PDF

$$f_X(x) = \begin{cases} c(1 - x^2), & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of c.

(b) Find the CDF of X.

(c) Find $P(X < 0.5)$.

(d) Find $P(X > 0.75)$.

(e) Find $P(X > 0.75 \mid X > 0.5)$.

(f) Find $E[X]$.

(g) Find $E[X^2]$.

(h) Find $\mathrm{Var}(X)$.

(i) Determine whether the density assigns greater probability to values near 0 or values near 1. Explain how this is reflected in the mean.

Part IV Integrated Challenge Problems

Challenge 24: Bayesian Decision-Making with Unequal Costs

A fraud-detection system examines online transactions. The probability that a transaction is fraudulent is

$$P(F) = 0.005.$$

The system raises an alert A with probabilities

$$P(A \mid F) = 0.98, \qquad P(A \mid F^c) = 0.02.$$

If an alerted transaction is blocked, the cost is \$20 when the transaction is legitimate and \$5 when the transaction is fraudulent. If an alerted transaction is allowed, the cost is \$1,000 when it is fraudulent and \$0 when it is legitimate.

(a) Find $P(A)$.
(b) Find $P(F \mid A)$.
(c) Find the conditional expected cost of blocking an alerted transaction.
(d) Find the conditional expected cost of allowing an alerted transaction.
(e) Determine the cost-minimizing decision after an alert.
(f) Let $p = P(F \mid A)$. Derive the threshold value of p above which blocking has a smaller expected cost than allowing.

Challenge 25: Finding the Required Number of Trials

A backup operation succeeds independently with probability 0.12 on each attempt. Attempts continue until at least one success occurs or until a fixed maximum of n attempts has been reached.

(a) Find the probability of at least one success in n attempts.
(b) Set up an inequality that guarantees a success probability of at least 0.95.
(c) Solve the inequality using logarithms.
(d) Find the smallest integer n satisfying the requirement.
(e) Repeat the calculation if the required success probability is increased to 0.99.
(f) Discuss how the required number of attempts changes when the single-attempt success probability decreases.

Challenge 26: Independent Increments in a Poisson Process

Customer messages arrive according to a nonhomogeneous process with the following piecewise-constant rates:

$$\lambda(t) = \begin{cases} 4 \text{ per hour}, & 9{:}00\text{–}10{:}00, \\ 7 \text{ per hour}, & 10{:}00\text{–}12{:}00. \end{cases}$$

Assume that arrivals in disjoint time intervals are independent and that the number of arrivals in each interval is Poisson with parameter equal to the integrated rate over that interval.

(a) Find the probability of exactly 3 arrivals between 9:00 and 10:00.

(b) Find the probability of exactly 5 arrivals between 10:00 and 11:00.

(c) Find the probability of exactly 8 arrivals between 9:00 and 11:00.

(d) Find the probability of 2 arrivals between 9:00 and 10:00 and 6 arrivals between 10:00 and 11:00.

(e) Given that 8 arrivals occur between 9:00 and 11:00, find the probability that exactly 3 occurred between 9:00 and 10:00.

(f) Find the expected number of arrivals between 9:30 and 11:30.

(g) Explain where the independent-increments assumption is used.

Challenge 27: Inferring a Density from Its Mean

A continuous random variable X has PDF

$$f_X(x) = \begin{cases} a + bx, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

It is known that

$$E[X] = \frac{7}{12}.$$

(a) Use the normalization condition to obtain an equation involving a and b.

(b) Use the given mean to obtain a second equation.

(c) Find a and b.

(d) Verify that the resulting function is nonnegative on $[0, 1]$.

(e) Find the CDF of X.

(f) Find the median of X.

(g) Find $E[X^2]$ and $\mathrm{Var}(X)$.

Challenge 28: A Mixture of Exponential Response Times

A request is sent to Server 1 with probability 0.70 and to Server 2 with probability 0.30. Conditional on the selected server, the response time is exponential:

$$X \mid S_1 \sim \text{Exponential}(0.5), \qquad X \mid S_2 \sim \text{Exponential}(0.2),$$

where rates are measured per second.

(a) Find $P(X > 4 \mid S_1)$ and $P(X > 4 \mid S_2)$.

(b) Find $P(X > 4)$.

(c) Given that $X > 4$, find $P(S_2 \mid X > 4)$.

(d) Find $E[X]$ using conditional expectation.

(e) Find the mixture CDF $F_X(x)$.

(f) Find the mixture PDF $f_X(x)$.

(g) Determine whether the unconditional distribution of X is exponential. Justify your answer.

(h) Discuss which server is more likely to have handled a request with an unusually long response time.

Challenge 29: Designing a Two-Sided Normal Alarm Threshold

Under normal operating conditions, a sensor reading X follows

$$X \sim N(100, 5^2).$$

An alarm is raised when

$$|X - 100| > c.$$

(a) Find c so that the false-alarm probability is 0.05.

(b) Express the alarm region as two numerical intervals.

(c) Suppose a fault changes the distribution to

$$X \sim N(112, 5^2).$$

Find the probability that the alarm detects the fault.

(d) Repeat part (c) if the fault distribution is

$$X \sim N(105, 5^2).$$

(e) Find c if the false-alarm probability is reduced to 0.01.

(f) Explain the trade-off between false-alarm probability and detection probability.

Challenge 30: Repeated Testing for a Rare Condition

A condition occurs in 1% of the population. A diagnostic test has sensitivity 0.95 and false-positive probability 0.05:

$$P(+ \mid D) = 0.95, \qquad P(+ \mid D^c) = 0.05.$$

A person is tested three times. Assume that the test results are conditionally independent given the person's true condition.

An alarm is raised if at least two of the three tests are positive.

(a) Find the probability that the alarm is raised given that the person has the condition.

(b) Find the probability that the alarm is raised given that the person does not have the condition.

(c) Find the overall probability that the alarm is raised.

(d) Given that the alarm is raised, find the probability that the person has the condition.

(e) Find the false-negative probability of the three-test alarm rule.

(f) Given one positive result from a single test, find the probability that the person has the condition.

(g) Compare the posterior probability from one positive result with the posterior probability from the three-test alarm rule.

(h) Explain why a test with high sensitivity and specificity may still have a modest positive predictive value when the condition is rare.

(i) Discuss one advantage and one disadvantage of requiring at least two positive results before raising an alarm.

Formula Reference

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Bayes' theorem

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)}.$$

Expectation and variance

$$E[X] = \sum_x x\, p_X(x) \qquad \text{or} \qquad E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx,$$

$$\operatorname{Var}(X) = E[X^2] - \big(E[X]\big)^2.$$

Binomial distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E[X] = np, \qquad \operatorname{Var}(X) = np(1-p).$$

Poisson distribution

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad E[X] = \operatorname{Var}(X) = \lambda.$$

Geometric distribution

$$P(X = k) = (1-p)^{k-1} p, \qquad E[X] = \frac{1}{p}, \qquad \operatorname{Var}(X) = \frac{1-p}{p^2}.$$

Uniform distribution

$$E[X] = \frac{a+b}{2}, \qquad \operatorname{Var}(X) = \frac{(b-a)^2}{12}.$$

Exponential distribution

$$f_X(x) = \lambda e^{-\lambda x}, \qquad F_X(x) = 1 - e^{-\lambda x}, \qquad x \geq 0,$$

$$E[X] = \frac{1}{\lambda}, \qquad \operatorname{Var}(X) = \frac{1}{\lambda^2}.$$

Linear transformation

$$E[aX + b] = aE[X] + b, \qquad \operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

End of Exercise Set